

Homework 1 (Theory)

Name: PATHRI VIDYA PRAVEEN

Roll No.: CS24BTECH11047

Exercise 1.1:

⇒ Consider binary classification with  $\mathcal{X} = [0, 1]$  and concept class (assume that ground truth uses 1 element from the class):

$$\mathcal{H} = \{h_w(x) = 1_{\{a \leq x \leq b\}} : 0 \leq a \leq b \leq 1\}$$

⇒ Given dataset  $\mathcal{S} = \{(x_1, y_1), \dots, (x_n, y_n)\}$

⇒ Let distinct  $x$ -values in the dataset be

$$z_1 < z_2 < \dots < z_m$$

$$\forall z_j, \text{ define } P_j = \#\{i : x_i = z_j, y_i = 1\}$$

$$N_j = \#\{i : x_i = z_j, y_i = 0\}$$

$$P = \sum_{j=1}^m P_j = \text{total no. of positive examples.}$$

Taking some interval  $I = [a, b]$ ,  $J(I) = \{j : z_j \in I\}$

Define some weights  $w_j = N_j - P_j$

$$W(J) = \sum_{j \in J} w_j, \quad W(\emptyset) = 0 \quad (\text{say})$$

⇒ For  $I = [a, b]$  interval,

$$\hat{R}(I) = \sum_{x_i \notin I} 1_{\{y_i = 1\}} + \sum_{x_i \in I} 1_{\{y_i = 0\}}$$

$$= \sum_{j \notin J(I)} P_j + \sum_{j \in J(I)} N_j$$

$$\sum_j P_j = P$$

(2)

$$\hat{R}(I) = P - \sum_{j \in J(I)} P_j + \sum_{j \in J(I)} N_j = P + W(J(I))$$

This minimizes empirical risk.

$$\Rightarrow \text{let } W^* = \min_{J \subseteq [m]} W(J)$$

$J$  is contiguous/empty  $\rightarrow$  defining set has this prop. so we give constraint

$$J^* = \{J : J \text{ is contiguous or empty ; } W(J) = W^*\}$$

$$\hat{\mathcal{H}}_{\text{ERM}}(X) = \bigcup_{J \in J^*} \{[a, b] : 0 \leq a \leq b \leq 1, J([a, b]) = J\}$$

$$[a, b] \text{ is an ERM iff } \sum_{j \in J : z_j \in [a, b]} w_j = W^*$$

$\Rightarrow$  For some non-empty block  $J = \{l, l+1, \dots, r\}$   
Intervals whose included set is exactly  $J$  are those with

$$z_{l-1} \leq a \leq z_l, \quad z_r \leq b < z_{r+1}$$

Say  $z_0 = -1, z_{m+1} = 2$  (some no. outside  $\{0, 1\}$ ,  $< 0$ , and  $> 1$ )

$$I_{l,r} = \{[a, b] : 0 \leq a \leq b \leq 1, z_{l-1} < a \leq z_l, z_r \leq b < z_{r+1}\}$$

For empty block,  $J = \emptyset$ , interval must contain no data point.  
It must lie entirely in 1 gap b/w consecutive distinct  $x$ -values.

$$I_\emptyset = \bigcup_{k=0}^m \{[a, b] : 0 \leq a \leq b \leq 1, z_k < a \leq b < z_{k+1}\}$$

$$z_0 = -1, z_{m+1} = 2$$

$$\Rightarrow \text{let } S_0 = 0, S_k = \sum_{j=1}^k w_j$$

Block  $J = \{l, \dots, r\}$  has weight  $W(J) = S_r - S_{l-1}$

$$W^* = \min(0, \min_{0 \leq l-1 < r \leq m} (S_r - S_{l-1}))$$

③

⇒ Minimizers are

$$J^* = \{ \emptyset \text{ if } W^* = 0 \} \cup \{ \{i+1, \dots, r\} : 0 \leq i < r \leq m, S_r - S_i = W^* \}$$

⇒ So, we can say, ERM is not unique in general

We have ~~above~~ set of ERM's

$$\bigcup_{J \in J^*} \{ [a, b] : J([a, b]) = J \}$$

$J^*$  is set of index blocks (contiguous sets of data points) that achieve minimal empirical error. ERM is not

unique as even for single optimal index block, boundaries  $a$  and  $b$  can vary continuously as long as they don't cross any data points; and also many optimal blocks can tie and we can have  $\infty$  distinct intervals with same min. error

⇒ ① Take 2 distinct points  $x_1 < x_2$ . All 4 labellings are possible.

$$(1, 1) \rightarrow [x_1, x_2]$$

$$(0, 0) \rightarrow [c, c], c \in (x_1, x_2)$$

$$(1, 0) \rightarrow [x_1, x_1]$$

$$(0, 1) \rightarrow [x_2, x_2]$$

Shatters 2 points

$$\boxed{VCdim(\mathcal{H}) \geq 2}$$

② Take 3 points  $x_1 < x_2 < x_3$ .

$(1, 0, 1)$  labelling not possible as interval can contain  $x_1$  and  $x_3$  only if  $x_2$  also present.

3 point set cannot be shattered.

$$\boxed{VCdim(\mathcal{H}) \leq 2}$$

③  $\boxed{VCdim(\mathcal{H}) = 2}$



## Exercise 1.2:

④

⇒ Set of polynomial classifiers for regression under mean squared error:

$$\mathcal{H}_m = \left\{ h_w(x) : \sum_{i=0}^m a_i x^i : \underline{a} \in \mathbb{R}^{m+1} \right\}$$

$$1 \leq m \leq 10$$

⇒ let training set be  $S = \{(x_1, y_1), \dots, (x_n, y_n)\}$

⇒ For fixed  $m$ ,

$$\phi(x) = \begin{bmatrix} 1 \\ x \\ x^2 \\ \vdots \\ x^m \end{bmatrix} \in \mathbb{R}^{m+1}$$

$$a = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_m \end{bmatrix}$$

$$h_a(x) = \sum_{k=0}^m a_k x^k = \phi(x)^T a$$

$$\text{let } X = \begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_2 & x_2^2 & \dots & x_2^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^m \end{bmatrix} \in \mathbb{R}^{n \times (m+1)}$$

$$\text{let } y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$\Rightarrow \text{Empirical risk} = \hat{R}_S(a) = \frac{1}{n} \sum_{i=1}^n (h_a(x_i) - y_i)^2$$

$$\hat{R}_S(a) = \frac{1}{n} \|Xa - y\|^2$$

$$\text{Min } \hat{R}_S(a) \iff \text{Min } \|Xa - y\|^2$$

$$\Rightarrow \|Xa - y\|^2 = a^T X^T X a - 2a^T X^T y + y^T y$$

$$\Rightarrow \frac{\partial}{\partial a} \hat{R}_S(a) = \nabla_a \hat{R}_S(a) = \frac{2}{n} (X^T X a - X^T y)$$

$$\text{Set } \frac{\partial}{\partial a} \hat{R}_S(a) = 0 \longrightarrow X^T X a = X^T y$$

(5)

⇒ If  $X^T X$  is invertible, unique ERM is

$$a^* = (X^T X)^{-1} X^T y \quad \text{--- ①}$$

ERM polynomial =  $h^*(x)$

$$h^*(x) = \phi(x)^T a^* = \phi(x)^T (X^T X)^{-1} X^T y$$

$$h^*(x) = \sum_{k=0}^m a_k^* x^k \quad ; \quad a^* = (a_0^*, \dots, a_m^*)^T$$

This  $a^*$  solves

$$\sum_{j=0}^m \left( \sum_{i=1}^n x_i^{k+j} \right) a_j = \sum_{i=1}^n x_i^k y_i, \quad k=0 \text{ to } m$$

⇒ If  $X^T X$  is singular, ERM is not unique.

$$X^T X a = X^T y \quad \rightarrow \quad a \text{ is ERM.}$$

A closed form sol<sup>n</sup> for  $a$  is

$$a^* = \underbrace{(X^T X)^+}_{\text{Moore-Penrose pseudoinverse}} X^T y \quad \text{--- ②}$$

⇒ Closed form ERM expression for any fixed  $m$   
 ————— Eq<sup>s</sup> ① and ②

### Exercise 1.3

⇒ Multi-class classification problem with  $C$  classes.

For single training sample  $(x, y)$

$x \in \mathbb{R}^d$  — input vector

$y \in \{0, 1\}^C$  — one hot encoding target label.

$$\text{loss } L = - \sum_{k=1}^C y_k \log(\hat{y}_k)$$

$\hat{y} = \text{softmax}(z)$ ;  $z \in \mathbb{R}^C$  — pre-activation vector at output

$$\frac{\partial L}{\partial z} = \hat{y} - y \quad (\text{elementwise})$$

⇒ Perceptron: Single layer

Forward pass  $\rightarrow z = Wx + b$ ,  $\hat{y} = \text{softmax}(z)$   
 $W \in \mathbb{R}^{c \times d}$ ,  $b \in \mathbb{R}^c$

Backprop: Output error =  $\delta = \frac{\partial L}{\partial z} = \hat{y} - y \in \mathbb{R}^c$   
 $\delta = \hat{y} - y$  (weight)

Chain rule

$$\frac{\partial L}{\partial W} = \delta x^T$$

$$\frac{\partial L}{\partial W_{kj}} = \delta_k x_j$$

$$\frac{\partial L}{\partial b} = \delta$$

$$\frac{\partial L}{\partial b_k} = \delta_k \quad (\text{biases})$$

⇒ Fully connected network with 1 hidden layer:

$H$  = no. of hidden neurons = 20

Parameters:  $W_1 \in \mathbb{R}^{H \times d}$ ,  $b_1 \in \mathbb{R}^H$ ,  $W_2 \in \mathbb{R}^{c \times H}$ ,  $b_2 \in \mathbb{R}^c$

Forward pass:  $h = W_1 x + b_1$ ,  $a = \tanh(h)$ ,  $z = W_2 a + b_2$

$\hat{y} = \text{softmax}(z)$

Backprop:

Output layer  $\rightarrow$

$$\delta_2 = \frac{\partial L}{\partial z} = \hat{y} - y$$

$$\delta_2 = \hat{y} - y$$

$$\frac{\partial L}{\partial W_2} = \delta_2 a^T$$

$$\frac{\partial L}{\partial b_2} = \delta_2$$

Hidden layer  $\rightarrow$  propagating error backward

$$\frac{\partial L}{\partial a} = W_2^T \delta_2 \quad (z = W_2 a + b_2)$$

$$\left[ \frac{d}{dh} \tanh(h) = 1 - \tanh^2(h) = 1 - a^2 \right]$$

vector of 1's

$$\text{Hidden error} = \delta_1 = \frac{\partial L}{\partial h} = (W_2^T \delta_2) \odot (1 - a^2) = \delta_1$$

$$\frac{\partial L}{\partial W_1} = \delta_1 x^T$$

$$\frac{\partial L}{\partial b_1} = \delta_1$$



## Exercise 1.4!

⑦

⇒ Function

$$f(x_1, x_2, x_3) = \sin(x_1 x_2) + e^{x_1 - x_3} + x_2^2 - x_3$$

① ⇒ Gradient of  $f$  at  $[1, 1, 1] = x_0$

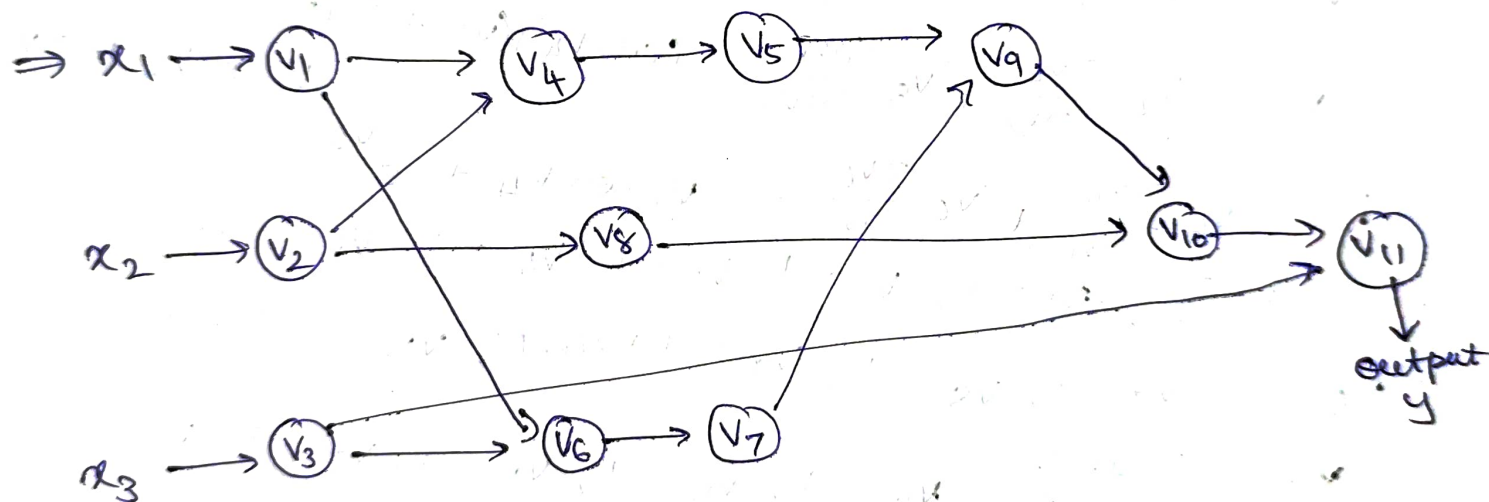
$$\Rightarrow \frac{\partial f}{\partial x_1} = x_2 \cos(x_1 x_2) + e^{x_1 - x_3} \Rightarrow \frac{\partial f}{\partial x_1} \Big|_{x=x_0} = \cos(1) + 1$$

$$\frac{\partial f}{\partial x_2} = x_1 \cos(x_1 x_2) + 2x_2 \Rightarrow \frac{\partial f}{\partial x_2} \Big|_{x=x_0} = \cos(1) + 2$$

$$\frac{\partial f}{\partial x_3} = -e^{x_1 - x_3} - 1 \Rightarrow \frac{\partial f}{\partial x_3} \Big|_{x=x_0} = -2$$

$$\nabla f(1, 1, 1) = \begin{bmatrix} 1 + \cos(1) \\ 2 + \cos(1) \\ -2 \end{bmatrix}$$

③ ⇒ Computation graph for  $f$



$$\Rightarrow v_1 = x_1 ; v_2 = x_2 ; v_3 = x_3$$

$$\Rightarrow v_4 = v_1 v_2 ; v_5 = \sin(v_4) ; v_6 = v_1 - v_3 ; v_7 = e^{v_6} ;$$

$$v_8 = v_2^2 ; v_9 = v_5 + v_7 ; v_{10} = v_9 + v_8 ;$$

$$v_{11} = v_{10} - v_3$$

$$\Rightarrow y = f(x_1, x_2, x_3) = v_{11}$$

④  $\Rightarrow \forall \text{ nodes } v_i, \text{ adjoint } = \bar{v}_i = \frac{\partial y}{\partial v_i} ; y = v_{11}$  ⑧

$\Rightarrow \bar{v}_{11} = 1$  Reverse mode autodiff

$\Rightarrow$  Traverse graph in reverse topological order

$$\bar{v}_{10} = \bar{v}_{11} \cdot \frac{\partial v_{11}}{\partial v_{10}} = 1 \cdot 1 = 1$$

$$\bar{v}_9 = \bar{v}_{10} \cdot \frac{\partial v_{10}}{\partial v_9} = 1 \cdot 1 = 1$$

$$\bar{v}_8 = \bar{v}_{10} \cdot \frac{\partial v_{10}}{\partial v_8} = 1 \cdot 1 = 1$$

$$\bar{v}_7 = \bar{v}_9 \cdot \frac{\partial v_9}{\partial v_7} = 1 \cdot 1 = 1$$

$$\bar{v}_6 = \bar{v}_7 \cdot \frac{\partial v_7}{\partial v_6} = 1 \cdot e^{v_6} = e^{v_6}$$

$$\bar{v}_5 = \bar{v}_9 \cdot \frac{\partial v_9}{\partial v_5} = 1 \cdot 1 = 1$$

$$\bar{v}_4 = \bar{v}_5 \cdot \frac{\partial v_5}{\partial v_4} = 1 \cdot \cos v_4 = \cos v_4$$

$$\bar{v}_3 = \bar{v}_6 \cdot \frac{\partial v_6}{\partial v_3} + \bar{v}_{11} \cdot \frac{\partial v_{11}}{\partial v_3} = (e^{v_6})(-1) + (1)(-1) = -e^{v_6} - 1$$

$$\bar{v}_2 = \bar{v}_4 \cdot \frac{\partial v_4}{\partial v_2} + \bar{v}_8 \cdot \frac{\partial v_8}{\partial v_2} = (\cos v_4)(v_1) + (1)(2v_2) = v_1 \cos v_4 + 2v_2$$

$$\bar{v}_1 = \bar{v}_4 \cdot \frac{\partial v_4}{\partial v_1} + \bar{v}_6 \cdot \frac{\partial v_6}{\partial v_1} = v_2 \cos v_4 + e^{v_6}$$

$\Rightarrow$  At  $[1, 1, 1]$ ,  $v_1 = 1, v_2 = 1, v_3 = 1, v_4 = 1, v_5 = \sin 1,$   
 $v_6 = 0, v_7 = 1, v_8 = 1, v_9 = 1 + \sin 1, v_{10} = 1 + \sin 1,$

$v_{11} = 1 + \sin 1.$

$\Rightarrow$  Adjoints at  $[1, 1, 1]$ :  $\bar{v}_{11} = 1, \bar{v}_{10} = 1, \bar{v}_9 = 1, \bar{v}_8 = 1, \bar{v}_7 = 1,$

$\bar{v}_6 = 1, \bar{v}_5 = 1, \bar{v}_4 = \cos 1, \bar{v}_3 = -2, \bar{v}_2 = 2 + \cos 1,$

$\bar{v}_1 = 1 + \cos 1$

$\Rightarrow$  Gradient from reverse mode autodiff

$$\nabla f(1, 1, 1) = \begin{bmatrix} \bar{v}_1 \\ \bar{v}_2 \\ \bar{v}_3 \end{bmatrix} = \begin{bmatrix} 1 + \cos 1 \\ 2 + \cos 1 \\ -2 \end{bmatrix}$$